Vidhur B Preman

Staff Engineer — WLAN & Bluetooth Coexistence

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SUMMARY

Embedded Firmware Engineer (7+ Years) specializing in Wi-Fi/Bluetooth firmware (Samsung Exynos) and automotive infotainment systems. Adept at cross-boundary debugging from hardware up to the Android framework.

- Core Stack: C/C++, Linux, Android AOSP (Driver/HAL/Framework), IEEE 802.11a-ax (STA/AP/NAN/P2P).
- Deep Debugging: Expert in RF interference, dual-radio coexistence, and protocol sniffers (Wireshark, btsnoop, Ellisys, Teledyne X500).
- Track Record: Shipped certified features (WFA, CarPlay, Android Auto) for global OEMs including VW, Mercedes-Benz, JLR, Tata, and Citroën.

CORE TECHNICAL SKILLS

Wi-Fi	IEEE 802.11a/b/g/n/ac/ax STA, AP, Wi-Fi NAN, Wi-Fi P2P/Wi-Fi Direct, WLAN-BT coexistence
Debug & Analysis	Wireshark, btsnoop, Ellisys, Teledyne X500, GDB (crash/stack-trace analysis), Trace32, Spectrum captures
Bluetooth & AOSP	Bluetooth Classic, BLE, LE Audio; Android AOSP Bluetooth stack, Framework/HAL integration; BlueZ/Linux Bluetooth exposure
Programming	C, C++, Python, Bash, Java
Platforms	Android AOSP, Android Automotive (AAOS) exposure, embedded Linux, RTOS, systemd
Tooling & CI/CD	Docker, Jenkins, JFrog Artifactory, Git, Jira, automated test/flashing frameworks

PROFESSIONAL EXPERIENCE

Staff Engineer – WLAN and BT Firmware

Feb 2023 – Present

Samsung Semiconductor India Research · Bengaluru

- Design and maintain the WLAN-BT coexistence subsystem on Samsung Exynos SoCs, including mailbox design, arbitration logic, and switch control logic between WLAN and BT radios, supporting Classic BT, BLE, and LE Audio use cases.
- Debug and root-cause cross-layer Bluetooth and Wi-Fi firmware issues using HCI/BT sniffer logs, GDB stack-trace analysis, and Trace32, working directly with BT stack and BT firmware teams to resolve customer-reported defects.
- Integrate and validate WLAN/BT drivers and middleware across multiple SoC platforms during silicon bring-up, using hardware debuggers, serial consoles, and flashing tools.
- Architect real-time coexistence state machines in C for dual-radio SoCs; design low-level embedded software for ARM Cortex-M microcontrollers, optimising RAM footprint and execution time for MAC-layer arbitration.
- Develop multi-protocol scheduling algorithms that dynamically prioritise radio access based on packet type, giving SCO/eSCO audio priority over background Wi-Fi traffic.
- Debug RF interference and timing-critical packet drops using Wi-Fi/Bluetooth protocol sniffers (Teledyne X500, Ellisys); analyse OTA performance in anechoic chambers to validate coexistence behaviour.
- Collaborate with hardware design teams across Wi-Fi and Bluetooth baseband chips; align firmware with IEEE 802.11 and Bluetooth Core specifications for compliance and interoperability.
- Develop predictive traffic-shaping firmware that aligns Wi-Fi traffic with BLE LE Audio connection events to prevent audio disruption.

- Design hardware-software co-design frameworks to offload critical timing-loop arbitration from the CPU to dedicated lower MAC hardware.
- Optimise power consumption profiles for mobile/IoT devices.
- Mentor engineers and collaborate daily across a large, globally distributed organisation spanning WLAN MAC, PHY, BT stack, BT firmware, and hardware design teams; author technical documentation and present findings to stakeholders.

Embedded Software Engineer – BT/WLAN Automotive Infotainment

Jul 2019 – Feb 2023

VisteonCorporation · Chennai

- Developed and integrated Bluetooth and Wi-Fi features for automotive infotainment systems on Android AOSP, including driver integration and HAL/framework-level bring-up on Qualcomm, NXP, and MediaTek SoCs.
- Authored high-level and detailed design documents, requirements sign-offs, and test plans; built Wi-Fi Access Point, Station, and Wi-Fi Direct features across 802.11n/ac/ax and WPA2/WPA3 security frameworks.
- Integrated Wi-Fi/BT coexistence mechanisms to prevent signal interference; implemented multi-profile and multi- device connectivity management, including dual-phone pairing.
- Optimised Bluetooth audio streaming quality using SBC, AAC, and aptX codecs; developed security pairing features implementing Secure Simple Pairing (SSP) and Out-of-Band (OOB) pairing.
- Worked directly with silicon vendors (Qualcomm, NXP) for chipset bring-up; ported and configured vendor-specific HCI drivers and firmware patches.
- Modified and extended the Android Bluetooth stack, Linux BlueZ, and OpenSynergy BT stack; implemented IPC mechanisms (Binder, DBus) to bridge the BT stack with vehicle applications.
- Optimised power management states including low-power sniff mode and BLE advertising intervals.
- Root-caused Bluetooth and Wi-Fi interoperability and coexistence issues across production and pre-production hardware using HCI logs, PCAP/sniffer logs, OTA captures, ADB, and Wireshark; automated test frameworks for regression coverage.
- Interfaced Bluetooth systems with vehicle networks using CAN, LIN, and Ethernet.
- Delivered and certified Bluetooth/Wi-Fi-dependent features including CarPlay, Android Auto, and Wi-Fi Alliance certification, partnering directly with Volkswagen, Mercedes-Benz, Jaguar Land Rover, Tata Motors, and Citroen engineering teams.
- Developed under ASPICE-compliant processes with MISRA C/C++ coding standards for safety-relevant firmware, complying with ISO 26262 (ASIL) functional safety requirements including watchdog and safe-state handling.
- Supported hardware-in-the-loop (HIL) testing and vehicle-level drive tests for validation.

RTL Design Intern

Mar 2019 – May 2019

FLDEC · India

- Designed and simulated RTL blocks using VHDL and Verilog with Xilinx Vivado, ModelSIM, and FPGA platforms.

EDUCATION

B.E. — Electronics and Communication Engineering

2015 – 2019 · CGPA 6.7/10

Veltech Engineering College, Anna University

- Final Year Project: Optimised Booth Radix-8 MAC Unit (VLSI) — patent applied; research paper published in IJTE Journal.
- Finalist, Eureka Hackathon, G.H. Raisoni Institute of Technology — Li-Fi based communication module for autonomous vehicles.
- Fire extinguishing drone demoed at Defence Expo, CVRDE, Avadi.
- NPTEL Certification: Programming, Data Structures and Algorithms using Python.